Customer Churn Intelligence in Banking

SECTOR: BANKING & FINANCIAL SERVICES

Team Details (Group 9)

MEMBERS

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FACULTY MENTOR

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Project Resources

GitHub Repository
github.com/lakshyabapna/SectionA_G9_Customer_Churn_Intelligence

Tableau Public Dashboard
https://public.tableau.com/views/DVA_Capstone_G9_1777464603860/Dashboard2?:language=enGB&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Submission Date
29/04/26

Context & Problem Statement

Sector Context

Highly competitive environment characterized by exceptionally low switching costs for customers.

Retention is significantly cheaper than acquisition, putting Customer Lifetime Value (CLV) heavily at risk.

Customer churn directly erodes core revenue, shrinks the deposit base, and stifles long-term growth.

Primary Decision-Makers

- CX / Retention Lead
- Product Head
- Analytics Manager

Core Problem Statement

> **"Which behavioral and financial factors drive churn, and how can high-risk customers be identified early enough for effective intervention?"**

Project Objectives

1. **Identify Key Drivers**
   Uncover hidden patterns in customer behavior and financials.
2. **Segment by Risk Level**
   Classify customers to prioritize retention resources effectively.
3. **Enable Proactive Retention**
   Deliver targeted interventions using data-backed insights.

Data Engineering (Source → Pipeline → Output)

Data Source

Dataset
Kaggle Banking Churn Prediction

Raw Volume
28,382 rows × 21 columns

Python ETL

Standardized columns & parsed transaction dates

Missing values: median for numeric, mode for categorical

Anomalies fixed: dependants > 10 replaced with median

Features & Output

New Derived Metrics
is_active, balance_diff, customer_tenure_years, age_group

Final Processed File
cleaned_banking_churn.csv
28,382 rows × 22 columns

Key Analytical Fields

`churn` `is_active` `current_balance` `previous_month_balance` `balance_diff` `customer_tenure_years` `age_group` `occupation` `gender`

KPI & Metrics Framework

Churn Rate

18.53%

Business Relevance: Measures the share of customers lost. Primary metric for evaluating retention success.

Active Customer Rate

88.64%

Business Relevance: Indicates engagement health. Low activity heavily correlates with elevated churn risk.

Average Tenure

5.73 Yrs

Business Relevance: Highlights lifecycle stage. New customers (0-2 years) show the highest vulnerability.

Average Balance

$7,380.55

Business Relevance: Proxy for financial engagement. Higher balances signal stronger retention and value.

Avg Balance Change

-$114.63

Business Relevance: Captures financial trajectory. Negative values act as powerful early warning signals.

Key Insights (EDA)

1

Inactivity Drives Attrition

Inactive customers churn far more than active users, making engagement a primary metric.

2

Early Lifecycle Risk

New customers (0–2 years) are most vulnerable, requiring stronger onboarding strategies.

3

Declining Balance as a Precursor

Negative balance difference strongly precedes churn, acting as a clear early warning signal.

4

High-Risk Segments Amplify Loss

Classified high-risk segments exhibit disproportionately elevated churn rates across metrics.

5

Age Correlates with Wealth Stability

Customers 60+ hold higher balances, while younger groups show higher financial variability.

6

Occupation Influences Retention

Churn patterns vary distinctly by occupation, with self-employed being the largest segment.

Statistical Validation

Chi-Square Test

ACTIVITY STATUS VS. CHURN

Highly Significant Result

Inactive customers are significantly more likely to churn.

The extremely low p-value statistically validates the EDA finding that engagement (recent transaction activity) is a primary driver of retention.

Independent T-Test

BALANCE CHANGE (CHURNED VS RETAINED)

Highly Significant Result

Churned customers show a significantly more negative balance change.

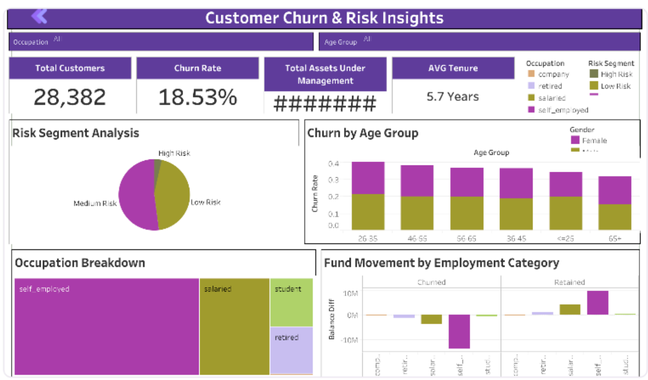
The substantial negative t-statistic confirms that declining balances reliably differentiate churning customers from retained customers.

Tableau Dashboard Walkthrough

Tableau Public URL: https://public.tableau.com/views/DVA_Capstone_G9_17774646013860/Dashboard2?:language=en-GB&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Interactive Filters: Gender | Age Group | Occupation | Risk Segment

1. Executive View

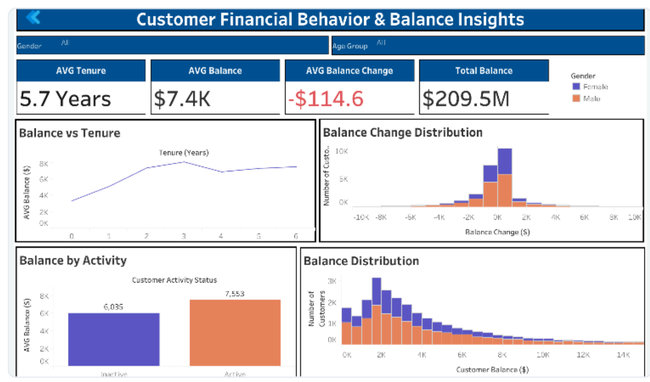


Core Overview & KPIs:

- **Total Customers:** 28,382
- **Churn Rate:** 18.53%
- **Active Rate:** 88.64%
- **Avg Balance:** $7,380.55

Purpose: High-level monitoring and quick drill down to at-risk cohorts.

2. Financial Insights

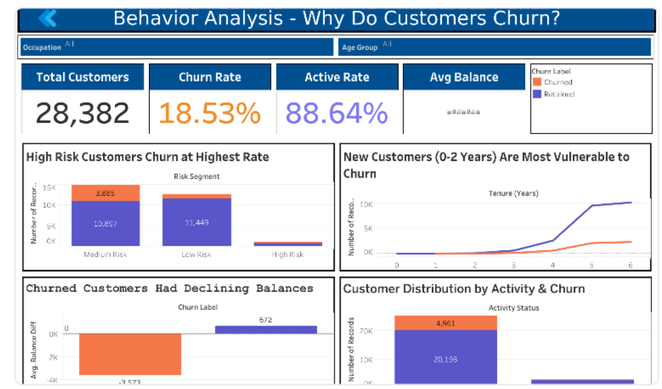


Key Visualizations:

- Balance vs. Tenure patterns
- Balance Change Distribution
- Overall Balance Distribution
- Avg balance by Activity Status

Purpose: Understand financial trajectory and engagement across segments.

3. Behavior Analysis



Actionable Insights:

- Inactivity vs. Churn correlation
- Balance Decline warning patterns
- Transactions comparison
- High-risk segment identification

Purpose: Identify behavioral drivers and trigger conditions for intervention.

Business Recommendations

1. Reactivate Inactive Users

Deploy targeted engagement nudges, fee waivers, and conditional rewards to re-engage dormant accounts.

2. Strengthen 0-2 Year Onboarding

Implement structured 30/60/90-day touchpoints and starter product offers to build early stickiness.

3. Monitor Balance Declines

Configure automated alerts for sustained negative balance differences to trigger pre-churn outreach.

4. Segment-Based Retention

Prioritize High and Medium risk cohorts with tailored campaigns rather than generic retention efforts.

5. Run on Dashboards

Establish weekly review rhythms using the Tableau dashboards to identify emerging risks and track intervention success.

Business Impact & Value

Early Churn Identification

Detect at-risk customers before they leave using key behavioral indicators like activity drops and balance declines.

Improved Retention Strategies

Deploy targeted, data-backed campaigns tailored to specific risk segments instead of generic marketing blasts.

Better Resource Allocation

Focus retention budgets and relationship manager bandwidth efficiently on high-risk and high-value cohorts.

Increased CLV

Extend customer lifespans, maximizing product depth, cross-sell opportunities, and overall long-term profitability.

Reduced Revenue Loss

Directly protect the bottom line by preventing the exit of profitable accounts and avoiding high acquisition costs.

Proactive Decision-Making

Shift the organization from reactive damage control to proactive, analytics-driven customer management.

Limitations & Next Steps

Limitations

Snapshot Dataset: Lacks customer-level time series data to observe micro-behavioral changes over time.

Limited Behavioral Features: Key signals like transaction frequency and digital channel usage are missing.

Data Accuracy: Churn labels are assumed accurate; median/mode imputation may introduce minor biases.

Outlier Handling: Winsorization of extremes might suppress valid but rare high-value or high-risk cases.

Future Enhancements

Machine Learning: Build a supervised predictive model (e.g., Random Forest) for individual-level churn propensity scoring.

Real-Time Integration: Connect Tableau dashboards to live CRM feeds for automated, real-time balance decline alerts.

Data Enrichment: Incorporate service complaints, product portfolio depth, and channel usage to refine risk segments.

A/B Testing & Survival Analysis: Design controlled experiments for reactivation offers and estimate optimal intervention timing.

Team & Contribution Matrix

Group 9 · Section A | Newton School of Technology | **Faculty Mentor:** Aayushi Mam

Team Member	Dataset & Sourcing	ETL & Cleaning	EDA & Analysis	Statistical Analysis	Tableau Dashboard	Report Writing	PPT & Viva
Lakshya Bapna 2401010252	✔	✔			✔		
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Project Verification & Traceability

All team contributions are fully verifiable via GitHub Insights, PR history, and committed files at:
`https://github.com/lakshyabapna/SectionA_G9_Customer_Churn_Intelligence`